

Heat-extension overdetermination from weighted Schiffer: computation and an obstruction

May 11, 2026

1 Goal

The companion note (`direction2-barrel-limit.tex`) proposed the following deliverable: show that a weighted Schiffer pair (Ω, V) forces its heat extension $h_{\Omega, V}$ (Definition 2.7 of de Dios Pont, arXiv:2412.06344) to be constant on the lateral parabolic boundary $\partial\Omega \times [0, 1]$, then test parabolic Serrin.

The result of this note is twofold:

- *First order* (the easy direction): the weighted Schiffer overdetermination propagates to first order in t , so $h \equiv c$ and $\partial_t h \equiv 0$ on $\partial\Omega \times \{0\}$.
- *Second order* (the obstruction): there is an explicit boundary expression $\eta_2|_{\partial\Omega}$ involving the Hessians of V and ϕ that does *not* vanish for a generic weighted Schiffer pair. Lateral parabolic Schiffer is therefore strictly stronger than stationary weighted Schiffer.

This kills the “tautological propagation” clause in the previous note and reshapes the strategy: lateral parabolic Schiffer is a finer problem than weighted Schiffer, and the gap is parametrised by the Hessian-level data $\nabla^2 V \cdot \nabla^2 \phi$ at $\partial\Omega$.

2 Setup

Let (Ω, V) be a smooth convex pair in $\mathbb{R}^n$: Ω bounded convex with smooth boundary, $V \in C^\infty(\overline{\Omega})$ convex. The Langevin operator

$$L_{\Omega, V} u := -\Delta u + \nabla V \cdot \nabla u$$

on the weighted space $L^2(e^{-V} dx)$ with Neumann boundary data has a discrete spectrum starting at 0 with first nonzero eigenvalue $\lambda := \lambda_{\Omega, V}$ and (up to sign) first nontrivial eigenfunction $\phi := \phi_{\Omega, V}$. Throughout, assume the *stationary weighted Schiffer condition*:

$$L_{\Omega, V} \phi = \lambda \phi \text{ in } \Omega, \quad \partial_\nu \phi = 0 \text{ on } \partial\Omega, \quad \phi|_{\partial\Omega} = c. \quad (1)$$

The heat extension $h := h_{\Omega, V}$, following Definition 2.7 of 2412.06344, is the bounded solution of

$$\partial_t h = \frac{1}{8} (\Delta_x h + \lambda h) \text{ on } \Omega \times (0, 1), \quad \partial_\nu h = 0 \text{ on } \partial\Omega \times (0, 1), \quad h(x, 0) = \phi(x). \quad (2)$$

Crucially, the Laplacian in (2) is the *unweighted* one. This is exactly what comes out of the barrel limit, because $F_d(\Omega, V) \subset \mathbb{R}^{n+d+1}$ carries the Euclidean structure.

3 Boundary identities at $t = 0$

We collect three identities that hold on $\partial\Omega$ under (1).

Lemma 1. *Under (1), on $\partial\Omega$ one has*

$$\phi = c, \quad \nabla\phi = 0, \quad \Delta\phi = -\lambda c.$$

Proof. The first is given. The second follows because $\phi|_{\partial\Omega} = c$ makes the tangential gradient $\nabla^T\phi$ vanish, while $\partial_\nu\phi = 0$ kills the normal component; hence the full gradient vanishes on $\partial\Omega$. For the third, evaluate (1) on $\partial\Omega$: $-\Delta\phi + \nabla V \cdot \nabla\phi = \lambda\phi$. With $\nabla\phi = 0$ this gives $\Delta\phi = -\lambda c$. $\square$

4 First-order propagation

Define the residues

$$\eta_k := (\Delta + \lambda)^k \phi, \quad k \geq 0.$$

By induction from (2),

$$\partial_t^k h(x, 0) = 8^{-k} \eta_k(x) \quad \text{in } \Omega, \quad k \geq 0. \quad (3)$$

The lateral parabolic Schiffer condition $h \equiv c$ on $\partial\Omega \times [0, 1]$ is equivalent (by real-analyticity of h in t , which holds because the equation is analytic with analytic data) to

$$\eta_k|_{\partial\Omega} \equiv \begin{cases} c & k = 0 \\ 0 & k \geq 1 \end{cases} \quad \text{and} \quad \partial_\nu \eta_k|_{\partial\Omega} \equiv 0 \text{ for all } k \geq 0. \quad (4)$$

Lemma 2 (First-order matching). $\eta_1 = \nabla V \cdot \nabla\phi$ in Ω . In particular $\eta_1|_{\partial\Omega} \equiv 0$.

Proof. From $L_{\Omega, V}\phi = \lambda\phi$, i.e. $-\Delta\phi + \nabla V \cdot \nabla\phi = \lambda\phi$, rearrange: $\Delta\phi = \nabla V \cdot \nabla\phi - \lambda\phi$. Hence $\eta_1 = (\Delta + \lambda)\phi = \nabla V \cdot \nabla\phi$. On $\partial\Omega$ apply Lemma 1. $\square$

Corollary 3. $\partial_t h(x, 0) = 0$ for $x \in \partial\Omega$. So the lateral parabolic Schiffer condition holds at order zero and one in t .

5 Second-order obstruction

Proposition 4 (Second-order residue). *On $\partial\Omega$ one has*

$$\eta_2 = 2 \operatorname{tr}(\nabla^2 V \cdot \nabla^2 \phi) + \langle \nabla^2 \phi \cdot \nabla V, \nabla V \rangle. \quad (5)$$

Proof. By Lemma 2, $\eta_1 = \nabla V \cdot \nabla\phi$. Carry out the Leibniz expansion. With summation convention,

$$\begin{aligned} \Delta(\partial_i V \partial_i \phi) &= (\Delta \partial_i V) \partial_i \phi + 2 \partial_j (\partial_i V) \partial_j (\partial_i \phi) + \partial_i V (\Delta \partial_i \phi) \\ &= (\partial_i \Delta V) \partial_i \phi + 2 \partial_i \partial_j V \partial_i \partial_j \phi + \partial_i V \partial_i (\Delta \phi). \end{aligned}$$

Using $\Delta\phi = \nabla V \cdot \nabla\phi - \lambda\phi$ gives $\partial_i(\Delta\phi) = \partial_i(\partial_j V \partial_j \phi) - \lambda \partial_i \phi = \partial_i \partial_j V \partial_j \phi + \partial_j V \partial_i \partial_j \phi - \lambda \partial_i \phi$, hence

$$\partial_i V \partial_i (\Delta \phi) = \partial_i V \partial_i \partial_j V \partial_j \phi + \partial_i V \partial_j V \partial_i \partial_j \phi - \lambda \partial_i V \partial_i \phi.$$

Restrict to $\partial\Omega$. By Lemma 1, $\nabla\phi = 0$ there, so every term carrying a single $\partial_i\phi$ vanishes:

$$(\partial_i\Delta V)\partial_i\phi|_{\partial\Omega} = 0, \quad \partial_i V \partial_i \partial_j V \partial_j \phi|_{\partial\Omega} = 0, \quad \lambda \partial_i V \partial_i \phi|_{\partial\Omega} = 0.$$

Hence

$$\Delta\eta_1|_{\partial\Omega} = 2 \partial_i \partial_j V \partial_i \partial_j \phi + \partial_i V \partial_j V \partial_i \partial_j \phi.$$

The term $\lambda\eta_1 = \lambda\nabla V \cdot \nabla\phi$ vanishes on $\partial\Omega$ (again $\nabla\phi = 0$). Therefore on $\partial\Omega$

$$\eta_2 = (\Delta + \lambda)\eta_1 = 2 \operatorname{tr}(\nabla^2 V \nabla^2 \phi) + (\nabla V)^\top \nabla^2 \phi (\nabla V),$$

as claimed. $\square$

Corollary 5 (The genuine obstruction). *The lateral parabolic Schiffer condition implies, in addition to (1), the boundary constraint*

$$2 \operatorname{tr}(\nabla^2 V \nabla^2 \phi) + (\nabla V)^\top \nabla^2 \phi (\nabla V) \equiv 0 \quad \text{on } \partial\Omega. \quad (6)$$

This is a genuinely new constraint: for any convex weighted Schiffer pair (Ω, V) with V not identically constant, (6) need not hold. Hence lateral parabolic Schiffer is strictly stronger than stationary weighted Schiffer.

Remark 6 (Sanity checks).

- **Disc**, $V \equiv 0$. Then $\eta_1 = \nabla V \cdot \nabla\phi = 0$ everywhere, hence $\eta_k \equiv 0$ for all $k \geq 1$, and lateral parabolic Schiffer holds with $h \equiv \phi$ stationary.
- **Disc, radial V , radial ϕ** . $\eta_1 = V'(r)\phi'(r)$. On ∂B ($r = R$), $\phi'(R) = \partial_\nu \phi = 0$, so $\eta_1|_{\partial B} = 0$. For η_2 on the boundary, $\nabla^2 V = V''(R)\partial_\nu \otimes \partial_\nu + r^{-1}V'(R)P_T$ and $\nabla^2 \phi = \phi''(R)\partial_\nu \otimes \partial_\nu + r^{-1}\phi'(R)P_T = \phi''(R)\partial_\nu \otimes \partial_\nu$, where P_T is the tangential projection. Then $\operatorname{tr}(\nabla^2 V \nabla^2 \phi) = V''(R)\phi''(R)$ and $(\nabla V)^\top \nabla^2 \phi (\nabla V) = V'(R)^2 \phi''(R)$, so (6) becomes $\phi''(R)(2V''(R) + V'(R)^2) = 0$. For generic radial convex V , this fails unless either $\phi''(R) = 0$ or $2V''(R) + V'(R)^2 = 0$, neither of which is automatic. Hence even the disc with a generic radial convex potential is a counterexample to “stationary weighted Schiffer $\Rightarrow$ lateral parabolic Schiffer.”

6 The cap obstruction (high-dim no-go)

The companion note observed (using Theorem 2.6 of 2412.06344) that a putative Schiffer counterexample of barrel form $F_d(\Omega, V)$ would force, by tracking the cap, $\phi(x) = c$ on all of Ω . We restate this for the record because it is the cleanest formal payoff of the barrel correspondence.

Proposition 7 (No barrel-lifted high-dim Schiffer counterexamples). *Let (Ω, V) be a smooth convex pair with a spectral gap and $\lambda_{\Omega, V} \leq 2$. Suppose the high-dimensional convex set $F_d(\Omega, V)$ admits a Neumann Laplace eigenfunction u_d such that $u_d \equiv c_d$ on $\partial F_d(\Omega, V)$ for a sequence of d , with $u_d/\|u_d\|$ converging to the ground eigenfunction along the subsequence. Then necessarily $\phi_{\Omega, V} \equiv c$ in Ω , hence $c = 0$ and $\phi_{\Omega, V} \equiv 0$, contradicting the non-triviality of $\phi_{\Omega, V}$.*

Proof. By Theorem 2.6(C.2)-(C.4) of 2412.06344, after the change of variables $t = 1 - 2|w|^2/d$, $u_d(x, w) = \psi_d(x, 1 - (|w|/\rho_d(V)(x))^2)$ and $\psi_d \rightarrow h_{\Omega, V}$ in C^0 on $\Omega \times [0, 1]$. The cap $\{|w| = \rho_d(V)(x)\}$ maps to $\Omega \times \{0\}$ and the lateral piece $\{x \in \partial\Omega, |w| < \rho_d(V)(x)\}$ maps to $\partial\Omega \times [0, 1]$. The Schiffer condition $u_d \equiv c_d$ on ∂F_d therefore passes to $h_{\Omega, V} \equiv c$ on $(\Omega \times \{0\}) \cup (\partial\Omega \times [0, 1])$. The cap clause $h_{\Omega, V}(x, 0) = \phi_{\Omega, V}(x) \equiv c$ forces $\phi_{\Omega, V}$ to be constant; but $\int_\Omega \phi_{\Omega, V} d\mu_{\Omega, V} = 0$ forces $c = 0$. $\square$

7 What the obstruction means for Direction 2

Three observations.

(i) Strategy B was too optimistic. The previous note claimed that stationary weighted Schiffer “tautologically” produces lateral parabolic Schiffer. Lemma 2 and Corollary 5 show that propagation holds only to first order; the second order introduces a Hessian-level constraint (6) that need not hold. Hence parabolic Serrin applied to the heat extension only delivers rigidity for the strict subclass of pairs (Ω, V) satisfying *both* stationary weighted Schiffer and the full tower (4).

(ii) Two productive reductions remain. The parabolic-Serrin angle is still alive in two forms:

- *Partial rigidity.* For the subclass of (Ω, V) at which all higher residues $\eta_k|_{\partial\Omega}$ happen to vanish, one obtains rigidity (provided parabolic Serrin fires for the relevant heat operator). The interesting question is how big this subclass is; (6) is the first selecting condition and is itself a sharp boundary identity that constrains (Ω, V) at the second Hessian order.
- *Distinguishing balls.* For the radial disc check in Remark 6, (6) reads $\phi''(R)(2V''(R) + V'(R)^2) = 0$. This is a one-parameter radial constraint on V at R . A natural follow-up is to ask whether *all* the constraints $\eta_k|_{\partial\Omega} = 0$ simultaneously force V to be one-dimensional ($V = V(r)$ in suitable coordinates) and Ω a ball.

(iii) An alternative geometric lift. The discrepancy between stationary weighted Schiffer and lateral parabolic Schiffer originates in one design choice: the barrel $F_d(\Omega, V)$ carries the Euclidean metric, so its Laplacian limits to $\Delta_x + \lambda$ (unweighted) rather than to $L_{\Omega, V} - \lambda$ (weighted). A warped variant

$$\tilde{F}_d(\Omega, V) := \Omega \times B_{\rho_d(V)}\mathbb{R}^{d+1} \text{ with metric } ds^2 = dx^2 + g(V(x), |w|) dw^2,$$

chosen so that the limit operator in t is the weighted heat operator $\partial_t = \frac{1}{8}(L_{\Omega, V} - \lambda)$, would make $\phi_{\Omega, V}$ stationary under the limit flow and lateral parabolic Schiffer would be automatic from stationary weighted Schiffer. This is the natural fix; the price is that $\tilde{F}_d(\Omega, V)$ is no longer a Euclidean convex set, and the high-dimensional “Schiffer for convex bodies” translation is lost.

8 Next steps

In decreasing order of priority:

1. **Explore the residue tower.** Compute $\eta_3|_{\partial\Omega}$ and check whether the residue tower can be summed: is the combination $h|_{\partial\Omega \times [0,1]} - c$ analytically representable in terms of $(\nabla V, \nabla^2 \phi)$ at the boundary?
2. **Resolve the radial test.** For the disc with radial convex V , determine the joint zero set of all residues $\{\eta_k|_{\partial B} = 0\}_{k \geq 1}$. The conjecture is that the only convex radial V with vanishing full tower is $V \equiv \text{const}$.
3. **Parabolic Serrin lookup.** For the heat operator $\partial_t - \frac{1}{8}(\Delta + \lambda)$ with overdetermined lateral data (constant Dirichlet AND Neumann), survey existing parabolic Serrin / overdetermined heat results (Magnanini–Sakaguchi, Vogel, Ciruolo–Magnanini–Vespri) and record exactly which of them apply.

4. **Warped barrel.** Attempt to construct the warped high-dimensional analog $\tilde{F}_d$ for which the limit operator is $L_{\Omega,V} - \lambda$ and lateral parabolic Schiffer is forced by stationary weighted Schiffer. Even if $\tilde{F}_d$ is not Euclidean-convex, it may produce an isoperimetric or weighted Brunn–Minkowski inequality that ports the barrel strategy.